

Операционные системы

Анализ файловой структуры UNIX. Команды для работы с файлами и каталогами

Дарья Калашникова

16 марта 2025

Российский университет дружбы народов, Москва, Россия

Цели и задачи работы

Ознакомление с файловой системой Linux, её структурой, именами и содержанием каталогов. Приобретение практических навыков по применению команд для работы с файлами и каталогами, по управлению процессами, по проверке использования диска и обслуживанию файловой системы.

- 1 Выполнить приимеры
- 2 Выполнить дествия по работе с каталогами и файлами
- 3 Выполнить действия с правами доступа
- 4 Получить дополнительные сведения при помощи справки по командам.

Процесс выполнения лабораторной работы

```
dvkashnikova@dvkashnikova:~$ cd
dvkashnikova@dvkashnikova:~$ touch abc1
dvkashnikova@dvkashnikova:~$ cp abc1 april
dvkashnikova@dvkashnikova:~$ cp abc1 may
dvkashnikova@dvkashnikova:~$ mkdir monthly
dvkashnikova@dvkashnikova:~$ cp april may monthly/
dvkashnikova@dvkashnikova:~$ cp monthly/may monthly/june
dvkashnikova@dvkashnikova:~$ ls monthly/
april  june  may
dvkashnikova@dvkashnikova:~$ mkdir monthly.00
dvkashnikova@dvkashnikova:~$ cp -r monthly monthly.00/
dvkashnikova@dvkashnikova:~$ cp -r monthly.00/ /tmp
dvkashnikova@dvkashnikova:~$
```



Рис. 1: Выполнение примеров

```
dvkashnikova@dvkashnikova:~$  
dvkashnikova@dvkashnikova:~$ cd  
dvkashnikova@dvkashnikova:~$ mv april july  
dvkashnikova@dvkashnikova:~$ mv july monthly.00/  
dvkashnikova@dvkashnikova:~$ ls monthly.00/  
july  monthly  
dvkashnikova@dvkashnikova:~$ mv monthly.00/ monthly.01  
dvkashnikova@dvkashnikova:~$ mkdir reports  
dvkashnikova@dvkashnikova:~$ mv monthly.01/ reports/  
dvkashnikova@dvkashnikova:~$ mv reports/monthly.01/ reports/monthly  
dvkashnikova@dvkashnikova:~$
```

Рис. 2: Выполнение примеров

```
dvkashnikova@dvkashnikova:~$  
dvkashnikova@dvkashnikova:~$ touch may  
dvkashnikova@dvkashnikova:~$ ls -l may  
-rw-r--r--. 1 dvkashnikova dvkashnikova 0 map 16 11:12 may  
dvkashnikova@dvkashnikova:~$ chmod u+x may  
dvkashnikova@dvkashnikova:~$ ls -l may  
-rwxr--r--. 1 dvkashnikova dvkashnikova 0 map 16 11:12 may  
dvkashnikova@dvkashnikova:~$ chmod u-x may  
dvkashnikova@dvkashnikova:~$ ls -l may  
-rw-r--r--. 1 dvkashnikova dvkashnikova 0 map 16 11:12 may  
dvkashnikova@dvkashnikova:~$ chmod g-r,o-r monthly/  
dvkashnikova@dvkashnikova:~$ chmod g+w abc1  
dvkashnikova@dvkashnikova:~$
```

Рис. 3: Выполнение примеров

Создание директорий и копирование файлов

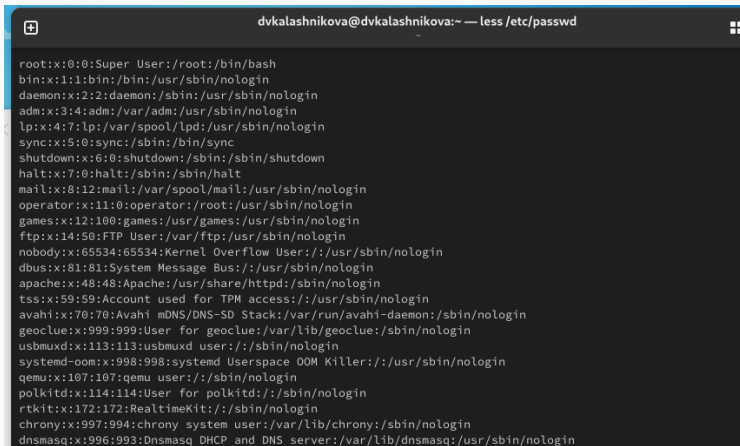
```
dvkashnikova@dvkashnikova:~$  
dvkashnikova@dvkashnikova:~$ cp /usr/include/linux/sysinfo.h ~  
dvkashnikova@dvkashnikova:~$ mv sysinfo.h equipment  
dvkashnikova@dvkashnikova:~$ mkdir ski.places  
dvkashnikova@dvkashnikova:~$ mv equipment ski.places/  
dvkashnikova@dvkashnikova:~$ mv ski.places/equipment ski.places/equiplist  
dvkashnikova@dvkashnikova:~$ cp abc1 ski.places/equiplist2  
dvkashnikova@dvkashnikova:~$ cd ski.places/  
dvkashnikova@dvkashnikova:~/ski.places$ mkdir equipment  
dvkashnikova@dvkashnikova:~/ski.places$ mv equiplist* equipment/  
dvkashnikova@dvkashnikova:~/ski.places$ ls  
equipment  
dvkashnikova@dvkashnikova:~/ski.places$ cd  
dvkashnikova@dvkashnikova:~$ mkdir newdir  
dvkashnikova@dvkashnikova:~$ mv newdir/ ski.places/  
dvkashnikova@dvkashnikova:~$ mv ski.places/newdir/ ski.places/plans  
dvkashnikova@dvkashnikova:~$
```

Рис. 4: Работа с каталогами

Работа с командой chmod

```
dvkashnikova@dvkashnikova:~$  
dvkashnikova@dvkashnikova:~$ mkdir australia play  
dvkashnikova@dvkashnikova:~$ touch my_os feathers  
dvkashnikova@dvkashnikova:~$ chmod 744 australia/  
dvkashnikova@dvkashnikova:~$ chmod 711 play/  
dvkashnikova@dvkashnikova:~$ chmod 544 my_os  
dvkashnikova@dvkashnikova:~$ chmod 664 feathers  
dvkashnikova@dvkashnikova:~$ ls -l  
итого 0  
-rw-rw-r--. 1 dvkashnikova dvkashnikova 0 map 16 11:10 abc1  
drwxr--r--. 1 dvkashnikova dvkashnikova 0 map 16 11:17 australia  
-rw-rw-r--. 1 dvkashnikova dvkashnikova 0 map 16 11:18 feathers  
drwxr-xr-x. 1 dvkashnikova dvkashnikova 74 фев 28 11:54 git-extended  
-rw-r--r--. 1 dvkashnikova dvkashnikova 0 map 16 11:12 may  
drwx--x--x. 1 dvkashnikova dvkashnikova 24 map 16 11:10 monthly  
-r-xr--r--. 1 dvkashnikova dvkashnikova 0 map 16 11:18 my_os  
drwx--x--x. 1 dvkashnikova dvkashnikova 0 map 16 11:17 play  
drwxr-xr-x. 1 dvkashnikova dvkashnikova 14 map 16 11:12 reports  
drwxr-xr-x. 1 dvkashnikova dvkashnikova 28 map 16 11:16 ski.places  
drwxr-xr-x. 1 dvkashnikova dvkashnikova 10 фев 28 11:38 work  
drwxr-xr-x. 1 dvkashnikova dvkashnikova 0 фев 28 11:28 Видео  
drwxr-xr-x. 1 dvkashnikova dvkashnikova 0 фев 28 11:28 Документы  
drwxr-xr-x. 1 dvkashnikova dvkashnikova 58 мар 6 18:07 Загрузки  
drwxr-xr-x. 1 dvkashnikova dvkashnikova 0 фев 28 11:28 Изображения  
drwxr-xr-x. 1 dvkashnikova dvkashnikova 0 фев 28 11:28 Музыка  
drwxr-xr-x. 1 dvkashnikova dvkashnikova 0 фев 28 11:28 Общедоступные  
drwxr-xr-x. 1 dvkashnikova dvkashnikova 0 фев 28 11:28 'Рабочий стол'  
drwxr-xr-x. 1 dvkashnikova dvkashnikova 0 фев 28 11:28 Шаблоны  
dvkashnikova@dvkashnikova:~$
```

Рис. 5: Настройка прав доступа



A terminal window with a dark background. The title bar at the top reads "dvkalashnikova@dvkalashnikova:~ — less /etc/passwd". The terminal displays the contents of the /etc/passwd file, showing system users and their associated shells and home directories. The users listed are: root, bin, daemon, adm, lp, sync, shutdown, halt, mail, operator, games, ftp, nobody, dbus, apache, tss, avahi, geoclue, usbmuxd, systemd-oom, qemu, polkitd, rtkit, chrony, and dnsmasq.

```
root:x:0:0:Super User:/root:/bin/bash
bin:x:1:1:bin:/bin:/usr/sbin/nologin
daemon:x:2:2:daemon:/sbin:/usr/sbin/nologin
adm:x:3:4:adm:/var/adm:/usr/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/usr/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/usr/sbin/nologin
operator:x:11:0:operator:/root:/usr/sbin/nologin
games:x:12:100:games:/usr/games:/usr/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/usr/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/usr/sbin/nologin
dbus:x:81:81:System Message Bus:/usr/sbin/nologin
apache:x:48:48:Apache:/usr/share/httpd:/sbin/nologin
tss:x:59:59:Account used for TPM access:/usr/sbin/nologin
avahi:x:70:70:Avahi mDNS/DNS-SD Stack:/var/run/avahi-daemon:/sbin/nologin
geoclue:x:999:999>User for geoclue:/var/lib/geoclue:/sbin/nologin
usbmuxd:x:113:113:usbmuxd user:/sbin/nologin
systemd-oom:x:998:998:systemd Userspace OOM Killer:/usr/sbin/nologin
qemu:x:107:107:qemu user:/sbin/nologin
polkitd:x:114:114>User for polkitd:/sbin/nologin
rtkit:x:172:172:RealtimeKit:/sbin/nologin
chrony:x:997:994:chrony system user:/var/lib/chrony:/sbin/nologin
dnsmasq:x:996:993:Dnsmasq DHCP and DNS server:/var/lib/dnsmasq:/usr/sbin/nologin
```

Рис. 6: Файл /etc/passwd

Работа с файлами и правами доступа

```
dvkashnikova@dvkashnikova:~$  
dvkashnikova@dvkashnikova:~$ cp feathers file.old  
dvkashnikova@dvkashnikova:~$ mv file.old play/  
dvkashnikova@dvkashnikova:~$ mkdir fun  
dvkashnikova@dvkashnikova:~$ cp -R play/ fun/  
dvkashnikova@dvkashnikova:~$ mv fun/ play/games  
dvkashnikova@dvkashnikova:~$ chmod u-r feathers  
dvkashnikova@dvkashnikova:~$ cat feathers  
cat: feathers: Отказано в доступе  
dvkashnikova@dvkashnikova:~$ cp feathers feathers2  
cp: невозможно открыть 'feathers' для чтения: Отказано в доступе  
dvkashnikova@dvkashnikova:~$ chmod u+r feathers  
dvkashnikova@dvkashnikova:~$ chmod u-x play/  
dvkashnikova@dvkashnikova:~$ cd play/  
bash: cd: play/: Отказано в доступе  
dvkashnikova@dvkashnikova:~$ chmod u+x play/  
dvkashnikova@dvkashnikova:~$
```

Рис. 7: Работа с файлами и правами доступа

```
dvkalashnikova@dvkalashnikova:~ — man mount
MOUNT(8) System Administration MOUNT(8)

NAME
    mount - mount a filesystem

SYNOPSIS
    mount [-h|-V]

    mount [-l] [-t fstype]

    mount -a [-fFnrsvw] [-t fstype] [-O optlist]

    mount [-fnrsvw] [-o options] device|mountpoint

    mount [-fnrsvw] [-t fstype] [-o options] device mountpoint

    mount --bind|--rbind|--move olddir newdir

    mount --make-[shared|slave|private|unbindable|rshared|rslave|rprivate|runbindable] mountpoint

DESCRIPTION
    All files accessible in a Unix system are arranged in one big tree, the file hierarchy, rooted at /. These files can be spread out over several devices. The mount command serves to attach the filesystem found on some device to the big file tree. Conversely, the umount(8) command will detach it again. The filesystem is used to control how data is stored on the device or provided in a virtual way by network or other services.

    The standard form of the mount command is:

        mount -t type device dir

    This tells the kernel to attach the filesystem found on device (which is of type type) at the directory dir. The option -t type is optional. The mount command is usually able to detect a filesystem. The root permissions are necessary to mount a filesystem by default. See section "Non-superuser mounts" below for more details. The previous contents (if any) and owner and mode of dir become invisible, and as long as this filesystem remains mounted, the pathname dir refers to the root of the filesystem on device.

    If only the directory or the device is given, for example:

        mount /dir
```

```
dvkcalashnikova@dvkcalashnikova:~ — man fsck
FSCK(8) System Administration FSCK(8)

NAME
    fsck - check and repair a Linux filesystem

SYNOPSIS
    fsck [-lsAVRTMNP] [-r [fd]] [-C [fd]] [-t fstype] [filesystem...] [--] [fs-specific-options]

DESCRIPTION
    fsck is used to check and optionally repair one or more Linux filesystems. filesystem can be a device name
    (e.g., /dev/hdc1, /dev/sdb2), a mount point (e.g., /, /usr, /home), or a filesystem label or UUID specifier
    (e.g., UUID=8868abf6-88c5-4a83-98b8-bfc24057f7bd or LABEL=root). Normally, the fsck program will try to
    handle filesystems on different physical disk drives in parallel to reduce the total amount of time needed
    to check all of them.

    If no filesystems are specified on the command line, and the -A option is not specified, fsck will default
    to checking filesystems in /etc/fstab serially. This is equivalent to the -As options.

    The exit status returned by fsck is the sum of the following conditions:

    0
        No errors

    1
        Filesystem errors corrected

    2
        System should be rebooted

    4
        Filesystem errors left uncorrected

    8
        Operational error

    16
        Usage or syntax error

    32
        Manual page fsck(8) line 1 (press h for help or q to quit)
```

```
dvkcalashnikova@dvkcalashnikova:~ — man mkfs
MKFS(8)                                     System Administration      MKFS(8)

NAME
    mkfs - build a Linux filesystem

SYNOPSIS
    mkfs [options] [-t type] [fs-options] device [size]

DESCRIPTION
    This mkfs frontend is deprecated in favour of filesystem specific mkfs.<type> utils.

    mkfs is used to build a Linux filesystem on a device, usually a hard disk partition. The device argument is either the device name (e.g., /dev/hda1, /dev/sdb2), or a regular file that shall contain the filesystem. The size argument is the number of blocks to be used for the filesystem.

    The exit status returned by mkfs is 0 on success and 1 on failure.

    In actuality, mkfs is simply a front-end for the various filesystem builders (mkfs.<fstype>) available under Linux. The filesystem-specific builder is searched for via your PATH environment setting only. Please see the filesystem-specific builder manual pages for further details.

OPTIONS
    -t, --type type
        Specify the type of filesystem to be built. If not specified, the default filesystem type (currently ext2) is used.

    fs-options
        Filesystem-specific options to be passed to the real filesystem builder.

    -V, --verbose
        Produce verbose output, including all filesystem-specific commands that are executed. Specifying this option more than once inhibits execution of any filesystem-specific commands. This is really only useful for testing.

    -h, --help
        Display help text and exit.

    -V, --version
        Print version and exit. (Option -V will display version information only when it is the only parameter, Manual page mkfs(8) line 1 (press h for help or q to quit))
```

```
dvkcalashnikova@dvkcalashnikova:~ — man kill

KILL(1)                                User Commands                                KILL(1)

NAME
    kill - terminate a process

SYNOPSIS
    kill [-signal|-s signal|-p] [-q value] [-a] [--timeout milliseconds signal] [--] pid|name...

    kill -l [number] | -L

DESCRIPTION
    The command kill sends the specified signal to the specified processes or process groups.

    If no signal is specified, the TERM signal is sent. The default action for this signal is to terminate the process. This signal should be used in preference to the KILL signal (number 9), since a process may install a handler for the TERM signal in order to perform clean-up steps before terminating in an orderly fashion. If a process does not terminate after a TERM signal has been sent, then the KILL signal may be used; be aware that the latter signal cannot be caught, and so does not give the target process the opportunity to perform any clean-up before terminating.

    Most modern shells have a builtin kill command, with a usage rather similar to that of the command described here. The --all, --pid, and --queue options, and the possibility to specify processes by command name, are local extensions.

    If signal is 0, then no actual signal is sent, but error checking is still performed.

ARGUMENTS
    The list of processes to be signaled can be a mixture of names and PIDs.

    pid
        Each pid can be expressed in one of the following ways:

        n
            where n is larger than 0. The process with PID n is signaled.

        0
            All processes in the current process group are signaled.

        -1
            Manual page kill(1) line 1 (press h for help or q to quit)
```


Выводы по проделанной работе

В ходе данной работы мы ознакомились с файловой системой Linux, её структурой, именами и содержанием каталогов. Научились совершать базовые операции с файлами, управлять правами их доступа для пользователя и групп. Ознакомились с Анализом файловой системы. А также получили базовые навыки по проверке использования диска и обслуживанию файловой системы.